STA2002: Probability and Statistics II
Distribution-free C.I., Sign Test, Wilcoxon Test

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Introduction

- In this lecture we will discuss how order statistics can help us to construct distribution-free confidence interval for percentiles. We will also learn about sign test and also Wilcoxon test.

- **Suggested reading**: Chapter 7.5, 8.4 of the textbook.

Introduction

- Most of the confidence intervals we have learned so far require knowledge/assumption on the distribution of the underlying sample, e.g. normal distribution.
- Here we would learn **distribution-free** confidence intervals, which can be applied even without assuming specific distributions.
- Later we would focus on random sample $X_1, \ldots, X_n$ i.i.d. which is from a **continuous** distribution.

Intuitive Example 1

Example 1

- Suppose that we have a random sample $X_1, \ldots, X_5$.
- Then naturally, we have the order statistics $X_{(1)}, X_{(2)}, \ldots, X_{(5)}$.
- Our goal is to estimate the median $m = \pi_{0.5}$.

Solution.

- First, it is not difficult to construct a point estimator of m as $X_{(3)}$.
- Suppose we use the interval $(X_{(1)}, X_{(5)})$ as a confidence interval for m.

Intuitive Example 1 (Cont'd)

- We consider the probability

$$\begin{aligned} P(X_{(1)} < m < X_{(5)}) &= 1 - P(X_{(1)} \geq m) - P(X_{(5)} \leq m) \\ &= 1 - P(X_1 \geq m, X_2 \geq m, \ldots, X_5 \geq m) \\ &\quad - P(X_1 \leq m, X_2 \leq m, \ldots, X_5 \leq m) \\ &= 1 - \prod_{i=1}^{5} P(X_i \geq m) - \prod_{i=1}^{5} P(X_i \leq m) \\ &= 1 - 0.5^5 - 0.5^5 = \frac{15}{16} = 94\%. \end{aligned}$$

- In other words, $(X_{(1)}, X_{(5)})$ is a 94% C.I. for $m = \pi_{0.5}$.
- Once we have data, say $x_1 = 8.7$, $x_2 = 0.1$, $x_3 = 4.4$, $x_4 = 3.1$, $x_5 = 6.0$, the above 94% C.I. is $(0.1, 8.7)$. □

Generalized Results

- In general, if we have a random sample $X_1, \ldots, X_n$ from an unknown continuous distribution with median m, then

$$\begin{aligned} P(X_{(1)} < m < X_{(n)}) &= 1 - P(X_{(1)} \geq m) - P(X_{(n)} \leq m) \\ &= 1 - P(X_1 \geq m, X_2 \geq m, \ldots, X_n \geq m) \\ &\quad - P(X_1 \leq m, X_2 \leq m, \ldots, X_n \leq m) \\ &= 1 - \prod_{i=1}^{n} P(X_i \geq m) - \prod_{i=1}^{n} P(X_i \leq m) \\ &= 1 - \left(\frac{1}{2}\right)^n - \left(\frac{1}{2}\right)^n = 1 - \left(\frac{1}{2}\right)^{n-1}. \end{aligned}$$

- In other words, $(X_{(1)}, X_{(n)})$ is a $1 - \left(\frac{1}{2}\right)^{n-1}$ C.I. for m.

Intuitive Example 2

Example 2

Suppose now the sample size is $n = 16$. What is the confidence level of the random interval $(X_{(4)}, X_{(11)})$ for the median $m = \pi_{0.5}$?

Intuitive Example 2 (Cont'd)

- Let W be the number of X_i that are less than m.
- In terms of indicator functions, it is

$$W = \sum_{i=1}^{16} \mathbf{1}_{(X_i<m)} \sim Bin(16, 1/2).$$

- Consider the probability

$$\begin{aligned} & P(X_{(4)} < m < X_{(11)}) \\ =& P(\text{at least 4 and at most 10 of } X_1, \ldots, X_{16} < m) \\ =& P(W = 4, 5, \ldots, 10) \end{aligned}$$

- Since each X_is are i.i.d. and $P(X_i < m) = 0.5$.
- Thus we can calculate and get,

$$\begin{aligned}
&P(X_{(4)} < m < X_{(11)}) \\
=&P(W = 4, 5, \ldots, 10) \\
=&\sum_{k=4}^{10} \binom{16}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{16-k} \\
=&\sum_{k=4}^{10} \binom{16}{k} \left(\frac{1}{2}\right)^{16} \\
=&0.8843.
\end{aligned}$$

- In other words, $(X_{(4)}, X_{(11)})$ is a 88.43% C.I. for m. □

Remark. Checking the binomial table (Table II Appendix B in textbook) or with help of computer, the above probability can be obtained.

Generalized Results

- In general, suppose we have a random sample $X_1, \ldots, X_n$ from an unknown continuous distribution with median m.
- Let W be the number of X_i that are less than m.
- In terms of indicator functions, it is

$$W = \sum_{i=1}^{n} \mathbf{1}_{(X_i<m)} \sim Bin(n, 1/2).$$

- Then for $i < j$,

$$\begin{aligned} & P(X_{(i)} < m < X_{(j)}) \\ =& P(\text{at least i and at most j-1 of } X_1, \ldots, X_n < m) \\ =& P(W = i, i+1, \ldots, j-1) \end{aligned}$$

Generalized Results (Cont'd)

- Continue the calculation,

$$\begin{aligned} & P(X_{(i)} < m < X_{(j)}) \\ =& P(W = i, i+1, \ldots, j-1) \\ =& \sum_{k=i}^{j-1} \binom{n}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{n-k} \\ =& \sum_{k=i}^{j-1} \binom{n}{k} \left(\frac{1}{2}\right)^n \end{aligned}$$

- In other words, $(X_{(i)}, X_{(j)})$ is a $\sum_{k=i}^{j-1} \binom{n}{k} \left(\frac{1}{2}\right)^n$ C.I. for m.

Approximation by Using Central Limit Theorem

- When n is large, we may as well use Central Limit Theorem to approximate the distribution of W. (due to the limited coverage of binomial distribution table).

- Question: Suppose now the sample size is $n = 27$. What is the approximate confidence level of the random interval $(X_{(10)}, X_{(18)})$ for m?

Approximation by Using Central Limit Theorem

- Let W be the number of X_i that are less than m, where

$$W \sim Bin(27, 1/2),$$

with $E(W) = 13.5$, $Var(W) = 27/4$.

- By Central Limit Theorem,

$$W \overset{approx}{\sim} N(13.5, 27/4).$$

- Consider Why?

Approximation by Using Central Limit Theorem

- Now, consider

$$
\begin{aligned}
& P(X_{(10)} < m < X_{(18)}) \\
=& P(10 \leq W \leq 17) \\
=& P(9.5 < W < 17.5) \\
=& P\left(\frac{9.5 - 13.5}{\sqrt{27/4}} < \frac{W - 13.5}{\sqrt{27/4}} < \frac{17.5 - 13.5}{\sqrt{27/4}}\right) \\
\approx& P\left(\frac{9.5 - 13.5}{\sqrt{27/4}} < N(0,1) < \frac{17.5 - 13.5}{\sqrt{27/4}}\right) \\
=& 0.8764.
\end{aligned}
$$

Approximation by Using Central Limit Theorem

- Note that we apply the half-unit correction, recall Chapter 5.7 in the book.
- On the other hand, the exact value by using the binomial pmf is 0.8779, so the normal approximation is reasonably close, that is,

$$
\begin{aligned}
& P(X_{(10)} < m < X_{(18)}) \\
=& P(10 \leq W \leq 17) \\
=& \sum_{k=10}^{17} \binom{27}{k} \left(\frac{1}{2}\right)^{k} \left(\frac{1}{2}\right)^{27-k} \\
=& \sum_{k=10}^{17} \binom{27}{k} \left(\frac{1}{2}\right)^{n} = 0.8779.
\end{aligned}
$$

C.I. for for Other π_p, e.g., $\pi_{0.25}$

- Instead of median $m = \pi_{0.5}$, we can apply the same technique to construct C.I. for any other percentiles π_p.
- Suppose now $n = 25$, and we would like to estimate $\pi_{0.25}$.
- Since $i = p(n+1) = 0.25(26) = 6.5$, we propose the point estimator of $\pi_{0.25}$ as,

$$\hat{\pi}_{0.25} = \frac{1}{2}X_{(6)} + \frac{1}{2}X_{(7)}.$$

- Next we continue to construct confidence interval for π_p which gives us interval estimation at a significance level.

C.I. for for Other π_p, e.g., $\pi_{0.25}$

- Let W be the number of X_i that are less than $\pi_{0.25}$.
- In terms of indicator functions, it is

$$W = \sum_{i=1}^{25} \mathbf{1}_{(X_i < \pi_{0.25})} \sim Bin(25, 0.25),$$

since X_is are i.i.d. and $P(X_i < \pi_{0.25}) = 0.25$.

- Next, we try to calculate the probability

$$P(X_{(3)} < \pi_{0.25} < X_{(10)})$$

- Continue the calculation,

$$
\begin{aligned}
& P(X_{(3)} < \pi_{0.25} < X_{(10)}) \\
=& P(\text{at least 3 and at most 9 of } X_1, \ldots, X_{25} < \pi_{0.25}) \\
=& P(W = 3, 4, \ldots, 9) \\
=& \sum_{k=3}^{9} \binom{25}{k} (0.25)^k (0.75)^{25-k}.
\end{aligned}
$$

- We can compute this probability by looking up in the binomial distribution table or by computer. Or alternatively, we can apply central limit theorem to approximate the probability value.

- Using the half-unit correction, we see that

$$
\begin{aligned}
& P(X_{(3)} < \pi_{0.25} < X_{(10)}) \\
=& P(3 \leq W \leq 9) \\
=& P(2.5 < W < 9.5) \\
\approx& \Phi\left(\frac{9.5 - \frac{1}{4} \cdot 25}{\sqrt{\frac{1}{4} \cdot \frac{3}{4} \cdot 25}}\right) - \Phi\left(\frac{2.5 - \frac{1}{4} \cdot 25}{\sqrt{\frac{1}{4} \cdot \frac{3}{4} \cdot 25}}\right) \\
=& 0.8917.
\end{aligned}
$$

- In other words, once we have data, the realized interval $(x_{(3)}, x_{(10)})$ is an approximate 89.17% C.I. for $\pi_{0.25}$.

Generalized Results

- We would like to estimate π_p for $p \in [0, 1]$.
- Let W be the number of X_i that are less than π_p.
- In terms of indicator functions, it is

$$W = \sum_{i=1}^{n} \mathbf{1}_{(X_i < \pi_p)} \sim Bin(n, p),$$

since each X_i are i.i.d. and $P(X_i < \pi_p) = p$.

Generalized Results (Cont'd)

- Consider the probability, for $i < j$,

$$\begin{aligned} & P(X_{(i)} < \pi_p < X_{(j)}) \\ =& P(\text{at least i and at most j-1 of } X_1, \ldots, X_n < \pi_p) \\ =& P(W = i, i+1, \ldots, j-1) \\ =& \sum_{k=i}^{j-1} \binom{n}{k} (p)^k (1-p)^{n-k}. \end{aligned}$$

- Again, compute this probability by looking up in the binomial distribution table or by computer. Or alternatively, we can apply central limit theorem to approximate the probability value.

Distribution-free Hypothesis Test

- Next, we will learn two distribution-free hypothesis tests:

1. Sign test
2. Signed rank test (Wilcoxon test), under the assumption that the distribution is symmetric and continuous

Sign Test

- Suppose we have a random sample $X_1, \ldots, X_n$, i.i.d., from a continuous distribution with median m.
- We would like to test for the median m:

$$H_0 : m = m_0$$

against either

$$H_1 : m > m_0$$

or

$$H_1 : m < m_0$$

or

$$H_1 : m \neq m_0.$$

- Sign test **counts the number of X_i's that favor H_1.**

Example 8.4-1

Example 3

- We would like to test,

$$H_0 : m = 6.2, \quad H_1 : m < 6.2.$$

- We have the following data:

6.8	5.7	6.9	5.3	4.1	9.8	1.7	7.0
2.1	19.0	18.9	16.9	10.4	44.1	2.9	2.4
4.8	18.9	4.8	7.9				

Example 8.4-1 (Cont'd)

- Define

$$N^- = \text{number of } X_i < 6.2$$

- Under H_0, m_0 is the median of the sample and so $N^- \sim Bin(20, 0.5)$.
- Back to the data we have, the realized value is

$$n^- = 9.$$

- Large value of N^- favours H_1, so the p-value is

$$\text{p-value} = P(N^- \geq 9; H_0) = P(Bin(20, 0.5) \geq 9) = 0.748$$

and we fail to reject H_0 at 5%.

Extended Discussion

- Now, suppose that our test is

$$H_0 : m = 7.2, \quad H_1 : m \neq 7.2.$$

- Define

$$N^- := \text{number of } X_i < 7.2$$
$$N^+ := \text{number of } X_i > 7.2$$

and we use the critical region

$$C = \{N^- \geq 15 \text{ or } N^+ \geq 15\}$$

Extended Discussion

- Back to the data we have,

$$n^- = 12, \quad n^+ = 8,$$

  and so we fail to reject H_0.

- Under H_0, $N^- \sim Bin(20, 0.5)$ and $N^+ \sim Bin(20, 0.5)$.
- The significance level (probability of type I error) is

$$\begin{aligned} & P(N^- \geq 15; H_0) + P(N^+ \geq 15; H_0) \\ =& 2P(Bin(20, 0.5) \geq 15) \\ =& 0.04. \end{aligned}$$

Example 8.4-1

Example 8.4-1 Let X denote the length of time in seconds between two calls entering a call center. Let m be the unique median of this continuous-type distribution. We test the null hypothesis $H_0: m = 6.2$ against the alternative hypothesis $H_1: m < 6.2$. Table II in Appendix B tells us that if Y is the number of lengths of time between calls in a random sample of size 20 that are less than 6.2, then the critical region $C = \{y : y \geq 14\}$ has a significance level of $\alpha = 0.0577$. A random sample of size 20 yielded the following data:

6.8	5.7	6.9	5.3	4.1	9.8	1.7	7.0
2.1	19.0	18.9	16.9	10.4	44.1	2.9	2.4
4.8	18.9	4.8	7.9				

Since $y = 9$, the null hypothesis is not rejected. ■

Intuition of Wilcoxon Test

- Recall that, the test statistic of **sign test** counts the number of X_i's that support H_1.
- For example, we count the number of X_i that are bigger than m_0, if $H_1 : m > m_0$.
- One drawback of the sign test is that we did not take into account of the magnitude of the difference between X_i and m_0.
- The **Wilcoxon test** considers the magnitude of the difference between X_i and m_0.

Wilcoxon Test

- **Assumption:** $X_1, \ldots, X_n$ are i.i.d. random samples from a symmetric and continuous distribution with median (and mean) m.
- **Null Hypothesis**: $H_0 : m = m_0$, where m_0 is a given constant.
- Next we introduce the calculation procedures in **Wilcoxon Test**.

Wilcoxon Test

- Step 1: Rank the absolute values $|X_1 - m_0|, |X_2 - m_0|, \ldots, |X_n - m_0|$ in ascending order according to the magnitude, and denote the rank of $|X_k - m_0|$ by R_k for $k = 1, 2, \ldots, n$.
- Step 2: For each R_k, we associate the sign of the difference $X_k - m_0$:
  If $X_k - m_0 > 0$, we use R_k;
  If $X_k - m_0 < 0$, we use $-R_k$.
  We call these **signed ranks**.
- Step 3: Compute the **Wilcoxon signed rank statistic**

$$W := \text{sum of signed ranks.}$$

This W is the constructed test statistic.

Example 8.4-2

Example 4

Suppose the lengths of $n = 10$ sunfish are,

$$x_i: \ 5.0 \ \ 3.9 \ \ 5.2 \ \ 5.5 \ \ 2.8 \ \ 6.1 \ \ 6.4 \ \ 2.6 \ \ 1.7 \ \ 4.3$$

We shall test

$$H_0 : m = 3.7, \quad \text{v.s.} \quad H_1 : m > 3.7.$$

Solution. We first prepare the signed ranks.

Example 8.4-2 (Cont'd)

- Check the signed ranks:

$x_k - m_0$:	1.3,	0.2,	1.5,	1.8,	−0.9,	2.4,	2.7,	−1.1,	−2.0,	0.6
$\lvert x_k - m_0 \rvert$:	1.3,	0.2,	1.5,	1.8,	0.9,	2.4,	2.7,	1.1,	2.0,	0.6
Ranks:	5,	1,	6,	7,	3,	9,	10,	4,	8,	2
Signed Ranks:	5,	1,	6,	7,	−3,	9,	10,	−4,	−8,	2

- The Wilcoxon signed rank statistic is then,

$$w = 5 + 1 + 6 + 7 - 3 + 9 + 10 - 4 - 8 + 2 = 25.$$

- Question: What is the distribution of test statistic W under H_0?

Distribution of W under $H_0 : m = m_0$

- Next we analyze the distribution of W under H_0.
- We can actually express W as

↳ # of $x_i < \pi_{lp}$.

$$W = \sum_{k=1}^{n} kS_k,$$

where under $H_0 : m = m_0$,

$$S_k = \begin{cases} 1 & \text{with probability } 1/2, \\ -1 & \text{with probability } 1/2. \end{cases},$$

and they are independent. Note that $E(S_k) = 0$ and $Var(S_k) = 1$.

Distribution of W under $H_0 : m = m_0$

- Under $H_0 : m = m_0$,

$$E(W) = \sum_{k=1}^{n} kE(S_k) = 0,$$

$$Var(W) = \sum_{k=1}^{n} k^2 Var(S_k) = \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}.$$

- Next We apply and use the limiting distribution proposed by Lyapunov,

$$\frac{W - 0}{\sqrt{\frac{n(n+1)(2n+1)}{6}}} \overset{approx}{\sim} N(0, 1).$$

Normal Approximation

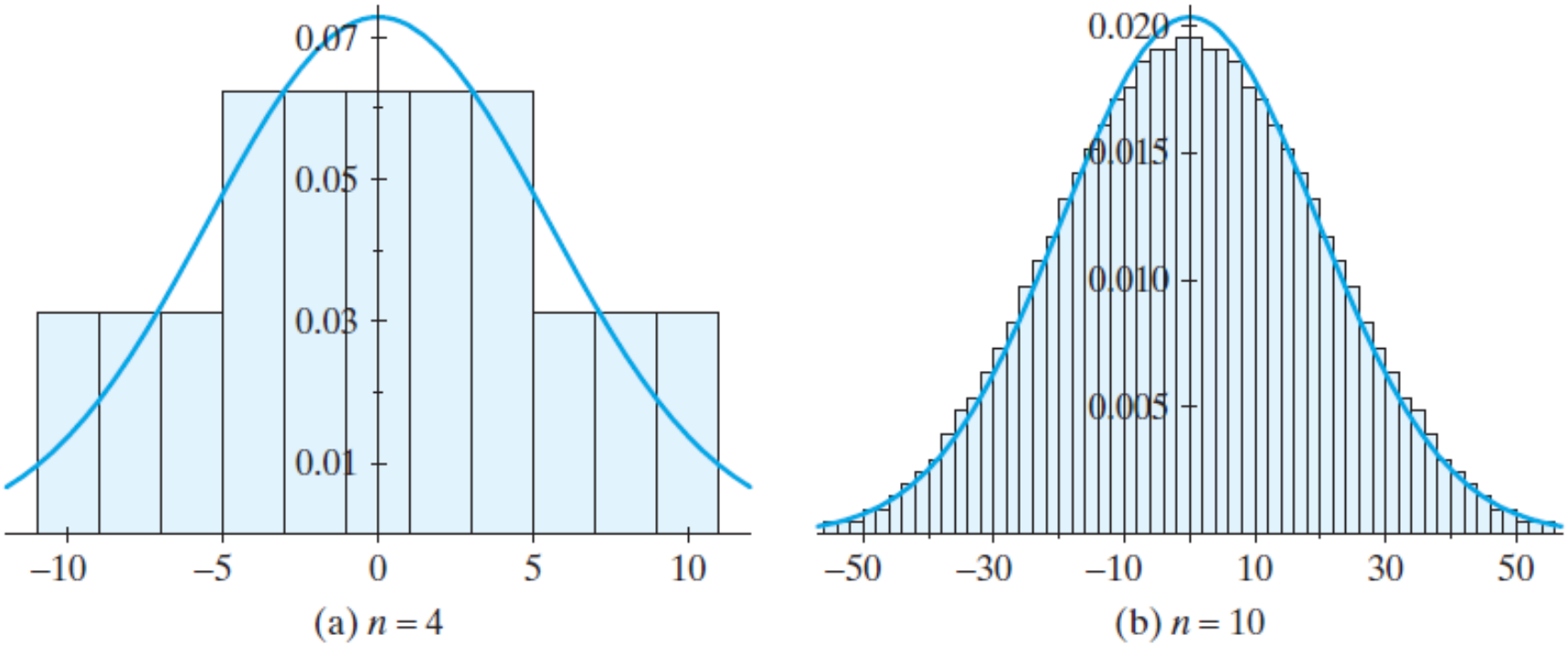


Figure 8.4-1 The Wilcoxon distribution

Example 8.4-3

Example 8.4-3

The moment-generating function of W or of V is given by

$$M(t) = \prod_{i=1}^{n} \frac{e^{-kt} + e^{kt}}{2}.$$

Using a computer algebra system such as *Maple*, we can expand $M(t)$ and find the coefficients of e^{kt}, which is equal to $P(W = k)$. In Figure 8.4-1, we have drawn a probability histogram for the distribution of W along with the approximating $N[0, n(n+1)(2n+1)/6]$ pdf for $n = 4$ (a poor approximation) and for $n = 10$. It is important to note that the widths of the rectangles in the probability histogram are equal to 2, so the "half-unit correction for continuity" mentioned in Section 5.7 now is equal to 1.

Continuity Correction

- Whole-unit continuity correction: Since possible values of W always differ by 2,

$$P(W \leq x) \approx \Phi\left(\frac{x+1}{\sqrt{\frac{n(n+1)(2n+1)}{6}}}\right).$$

- Now, under H_0, the significance level and p-value can be further calculated.

Example 8.4-2 continued

- $n = 10$, and so $n(n+1)(2n+1)/6 = 385$. The test is

$$H_0 : m = 3.7, \quad H_1 : m > 3.7.$$

- The Wilcoxon signed rank statistic is

$$w = 25.$$

- The p-value of the test is

$$P(W \geq 25; H_0) = 1 - P(W \leq 23; H_0) \approx 1 - \Phi\left(\frac{24}{\sqrt{385}}\right) = 0.11,$$

and we fail to reject H_0 at 5% significance. Note that the first equality above holds since W always differ by 2.

Data with ties

Under our assumption, the sample follows a continuous distribution, so theoretically we would not have the same data point appearing more than once. However, it does happen in practice. When there is a tie, we do the following:

- If $x_k = m_0$ for some k, delete these data and perform the test with a reduced sample size.
- If $|x_k - m_0| = |x_j - m_0|$ for some $k \neq j$, then each observation is assigned the average of the corresponding ranks.

Example 8.4-4

Let m be the median of a symmetric distribution of the continuous type. To test the hypothesis H_0: $m = 160$ against the alternative hypothesis H_1: $m > 160$, we take a random sample of size $n = 16$. For an approximate significance level of $\alpha = 0.05$, H_0 is rejected if the computed $W = w$ is such that

$$z = \frac{w}{\sqrt{16(17)(33)/6}} \geq 1.645,$$

or

$$w \geq 1.645\sqrt{\frac{16(17)(33)}{6}} = 63.626.$$

Example 8.4-4

Say the observed values of a random sample are 176.9, 158.3, 152.1, 158.8, 172.4, 169.8, 159.7, 162.7, 156.6, 174.5, 184.4, 165.2, 147.8, 177.8, 160.1, and 160.5. In Table 8.4-1, the magnitudes of the differences $|x_k - 160|$ have been ordered and ranked. Those differences $x_k - 160$ which were negative have been underlined, and the ranks are under the ordered values. For this set of data,

$$w = 1 - 2 + 3 - 4 - 5 + 6 + \cdots + 16 = 60.$$

Example 8.4-4

Table 8.4-1 Ordered absolute differences from 160

0.1	0.3	0.5	1.2	1.7	2.7	3.4	5.2
1	2	3	4	5	6	7	8
7.9	9.8	12.2	12.4	14.5	16.9	17.8	24.4
9	10	11	12	13	14	15	16

Example 8.4-4

Since $60 < 63.626$, H_0 is not rejected at the 0.05 significance level. It is interesting to note that H_0 would have been rejected at $\alpha = 0.10$, since, with a unit correction made for continuity, the approximate p-value is

$$
\begin{aligned}
p\text{-value} &= P(W \geq 60) \\
&= P\left(\frac{W-0}{\sqrt{(16)(17)(33)/6}} \geq \frac{59-0}{\sqrt{(16)(17)(33)/6}}\right) \\
&\approx P(Z \geq 1.525) = 0.0636.
\end{aligned}
$$

(*Maple* produces a p-value equal to $4{,}251/65{,}536 = 0.0649$.) Such a p-value would indicate that the data are too few to reject H_0, but if the pattern continues, we shall most certainly reject H_0 with a larger sample size. ■